University of Bahrain
Department of Computer Science
ITCS252 Discrete Structure I
RELATION EXAMPLES

Given $A = \{1, 6, 3, 9\}$. For each relation R on A state its properties and represent each relation by directed graph.

1. $R = \{(9, 9), (1, 7), (1, 1)\}$.

2. $R = \{(9, 9), (1, 3).(1, 1)\}$.

3. $R = \{(a, b) \in A^2 | a + b \leq 10\}$.

4. $R = \{(9, 9), (3, 3), (1, 1), (6, 6)\}$.

5. $R = \{(9, 9), (3, 3), (1, 1), (6, 6), (3, 1)\}$.

6. $R = \{(9, 9), (3, 3), (1, 1), (6, 6), (3, 1), (1, 3)\}$.

7. $R = \{(6, 9)\}$.

8. $R = \{(6, 9), (9, 6)\}$.

9. $R = \{(1,9),(6,9),(6,1)\}$.

10. $R = \{(1,9),(1,3)\}$.

11. $R = \{(1,1),(3,3),(3,1)\}$.

12. $R = \{(1,1),(3,3),(3,1),(1,3)\}$.

13. $R = \{(a,b) \in A^2 | a \mid b\}$.

14. $R = \{(a,b) \in A^2 | b \geq a\}$.

Given $A = \{0,1,2,4\}$. Give a relation S on A that satisfy the following properties. Draw directed graph For each relation.

1. S is reflexive symmetric, antisymmetric and transitive.

2. S is reflexive symmetric only .

3. S is antisymmetric and transitive only .

4. S is symmetric and transitive only .

5. S is not reflexive not symmetric, not antisymmetric and not transitive.

6. S is transitive.

7. S is symmetric, antisymmetric and not transitive.

8. S is symmetric only.

9. S is antisymmetric only .

show each relation properties. Explain

1. $\forall a, b \in Z : aRb \longleftrightarrow ab = 0.$

2. $\forall a, b \in Z : aRb \longleftrightarrow ab = 0.$

3. $\forall x, y \in Z : xRy \longleftrightarrow 7 \mid (x^2 - y^2).$

4. $\forall x, y \in Z^+ : xRy \longleftrightarrow xy \geq 0.$ vspace0.5cm

5. $\forall x, y \in Z^+ : xRy \longleftrightarrow xy \geq 0.$

6. $\forall x, y \in Z : aRb \longleftrightarrow x = \pm y.$

7. $\forall x, y \in Z : aRb \longleftrightarrow x + y = 0$.

8. $\forall x, y \in Z : aRb \longleftrightarrow x = 1$ or $y = 1$.

9. $\forall a, b \in Z : aRb \longleftrightarrow a \mid b.$

show each relation properties. Explain

1. a is father of b.

2. a is the son of b.

3. a GPA is higher than b GPA.

4. a is friend of b.

5. a set in front of b.

6. a set beside b.

7. a is not taller than b.

Given $A = \{1, 3, 2, 4, 5\}$ and $B = \{2, 3, 4, 12, 36, \}$

1. $\forall a, b \in A : aRB \longleftrightarrow 2|(a - b)$
   Show that R is equivalence. Draw its directed graph and list its classes.

2. $\forall a, b \in B : aRB \longleftrightarrow a|b$ Show that R is Partial order. Draw its directed graph and Hasse diagram .

3. $\forall a, b \in Z : aRB \longleftrightarrow a|b$. Show that R is Partial order.
   Draw Hasse diagram for $C = \{1, 5, 10, 15, 3, 4, 20, 16, 40\}$.

4. $\forall a, b \in Z : aRB \longleftrightarrow a \equiv bmod8$.
   Show that R equivalence and list the element of 2 classes .

1. Given the following Hasse diagram . Give the partila order relation R .

2. Given the following classes. Give the equivalence relation R.